

Take Your Pick: CALIBRATION LABS Or EQUIPMENT



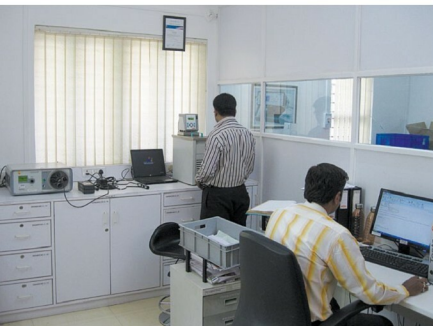
Saurabh Durgopal
is working as
technology
journalist at EFY

You would have noticed that the latest standards and technologies used by devices under test require extremely higher levels of accuracy than a decade ago. Since modern calibration equipment need to test and maintain these high levels of standards for test equipment that are already extremely accurate and precise, these need to deliver performance many times greater than the equipment being tested. Adith Vishwa, manager - repair and calibration, Rohde and Schwarz, explains, "Equipment required for calibration has to be up to four times better than original equipment."

"Calibration equipment is the ideal instrument preserved under standard ambient conditions in labs," says Abhijit Shah, assistant manager - service and calibration, Testo India Pvt Ltd. Comparing this with regular measurement instruments, "These experience several shocks and drifts in atmospheric aspects, as these are carried to several locations with different conditions," he adds.

Since calibration instruments are kept in their ideal state, these are more accurate

Fig. 1: Testo calibration lab



The need for calibration

B.C. Krishna, calibration lab and integrated service solutions manager, Keysight technologies India Pvt Ltd, explains the need for calibration as, "Calibration provides the confidence that a device's past performance was valid and that its future performance will continue to be good."

Different measurement parameters in the equipment drift slowly over time. Raviteja Chivukula, staff technical marketing engineer, National Instruments, adds, "This means that any measurement equipment has to be calibrated periodically for prolonged use."

Chivukula explains the different stages of calibration as:

Factory calibration. Done at the manufacturing site shortly after making the equipment

Shelf life. Time after factory calibration for which the instrument can be left unused without losing its calibration

Self-calibration. Done regularly, especially if temperature varies a lot from last-calibrated temperature

External calibration. Done over a period of one to two years

Re-calibration. Done whenever a need is felt

and precise as compared to measurement equipment. Vishwa adds, "Standard has to be maintained in these equipment, hence these are kept under stable environments."

T&M riding the smart wave

Other than maintaining optimum temperatures for highly sensitive calibration-standard equipment, these are now becoming smartdevices, too. Hiren Thaker, head - service, Rohde and Schwarz, explains, "The devices are now becoming intelligent." This is most certainly a welcome step towards the smartwave. "Equipment today have features for self-test and calibration," he adds.

This also makes the equipment interesting, as once the measurement begins to waver from the actual value, the device would either give an alarm so that proper measures can be taken, or self-calibration would bring the measured value close to the actual value.